A strong fixed-penalty baseline now works

The successful adaptive line search is our modification—not a reproduction of published Quiroga adaFGD.

The corrected sensing problem is recoverable. The remaining scientific tasks are to resolve the published step rule and complete a fair, independently tuned FRAMES comparison.

16 / 16

Modified FGD passes

One frozen setting; 8 calibration and 8 historical revalidation channels.

0 / 44

Printed-rule screen passes

14 finite completions; 30 numerical failures. These are 10k-budget trials.

1 / 2

Prior FRAMES passes

Both satisfy fidelity and TP; one misses the RMSE gate by 11.1%.

Headline scope: noiseless, rank-one, two-qubit recovery—not noisy or higher-rank tomography, not unseen-test validation, and not a demonstrated FRAMES advantage.

Positive higher-qubit context: earlier exploratory Armijo FGD also recovers four tested four-qubit channels. Its protocol differs from the new frozen-setting experiment.

Evidence: latest 64-run audited campaign [E1]; prior paired FRAMES diagnostic [E2].

What is controlled—and what is actually being compared

Same physical data and initial factor within each two-qubit channel; penalty policies remain method-specific.

Two qubits (d=4), rank-one Haar-unitary targets, 16 global input probes × 8 POVM outcomes = 128 Born probabilities. Full noiseless data. Normalized complex-uniform Gell-Mann coefficients initialize the physical factor; POVM constants a=b=0.1 are declared choices.

Fixed-penalty FGD: $J(U) = \frac{1}{2}\|\mathcal{A}(UU^\dagger) - y\|_2^2 + \lambda\|\mathrm{Tr}_{out}(UU^\dagger) - I\|_F^2$

Method label	Step-selection mechanism	TP policy
Printed Quiroga adaFGD	Residual / prediction norm ratio × fixed multiplier	Fixed λ; tuned independently
Armijo FGD control	Halve trial step until sufficient decrease	Fixed λ; separate control
Exact-ray FGD variant	Minimize quartic on positive factor ray; no backtracking	Fixed λ; separate modification
FRAMES	Stochastic FRAMES update and declared schedules	Fixed initial weight; increasing penalty

Joint recovery gate: fidelity ≥ 0.99 AND full-design RMSE ≤ 1e−6 AND normalized TP violation ≤ 1e−3, with a nonfailed terminal status. Normalized TP = ||Tr_out(UU†) − I||F / √d. The paper does not specify this numerical joint gate.

The paper gives an adaptive ratio, not this exact line search; unpublished constants prevent an exact implementation-level reproduction.

[P1] Quiroga & Kyrillidis, arXiv:2312.01311, §III-B and §V

Setup and conventions: qpt_quiroga_sensing.md; native archives [E1–E2]; published objective and step: [P1].

Latest result: one frozen setting passes all 16 channels

Exact factor-ray FGD, fixed $\alpha=5\times10^{-5}$ and $\lambda=128\alpha=0.0064$; selected before confirmation and revalidation.

Stage / channel seeds	Gate	RMSE range	Max. TP	Updates
Selected screen / 0–1	2/2	8.93e-11 – 1.00e-10	3.83e-10	1,401–1,882
Calibration confirmation / 2–7	6/6	7.73e-11 – 1.60e-10	5.54e-10	794–1,418
Historical revalidation / 100–107	8/8	4.83e-11 – 1.61e-10	7.80e-10	816–1,790

All 16 runs terminate by the full-gradient stopping test. Worst RMSE: 1.61×10^{-10}; worst normalized TP: 7.80×10^{-10}. Fidelity is approximately one (floating-point roundoff), in 794–1,882 accepted updates.

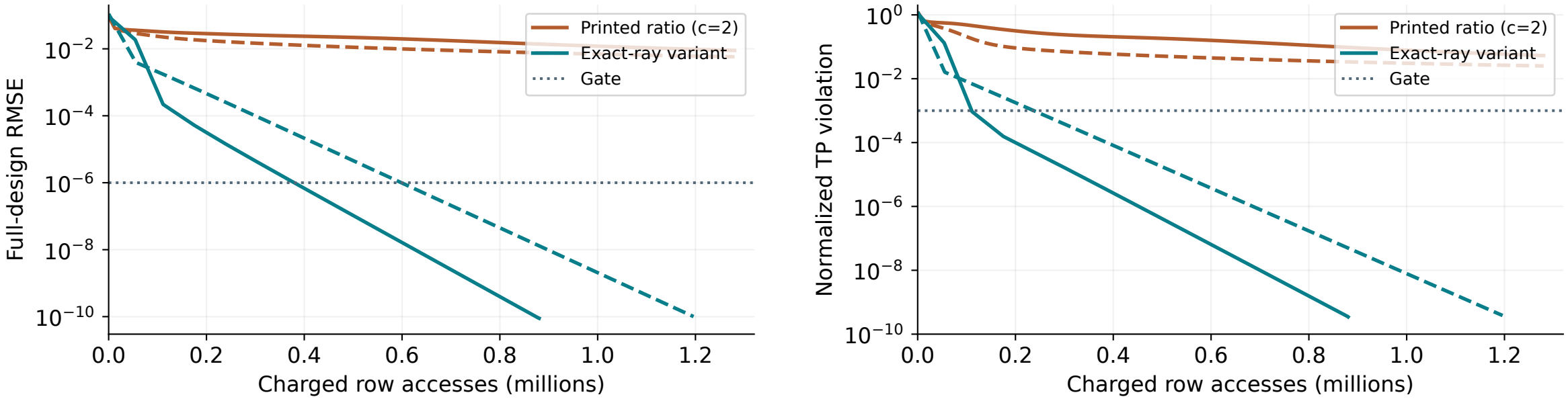
Selection used the worst-channel normalized gate deficit—not runtime. All three line-search penalties passed the two-channel screen; $\alpha=5\times10^{-4}$ actually used fewer charged rows than the selected $\alpha=5\times10^{-5}$. No claim of globally optimal hyperparameters.

Channels 100–107 were used in earlier campaigns: this is historical revalidation, not unseen validation. One initialization per channel. Final held-out channels 1000–1019 remain untouched.

[E1] audit_screen.json; audit_confirmation.json; audit_revalidation.json; selection_screen.json.

Step selection changes the observed convergence behavior

Paired channels 0/1: identical data, initialization, rank, objective and fixed λ=0.0064.



Printed-rule screen: 0/44 pass at 10,000 updates; best jointly stable new setting has RMSE 0.00578–0.00895. Larger tested multipliers often diverge. The shrinking-step tail supports a step-selection bottleneck on these cases; it is not a proof of nonconvergence.

Important context: four earlier printed-rule runs at 100,000 updates reached fidelity 0.9953–0.9987, but still failed both RMSE and TP gates. Row charges include line-search candidates but exclude spectral, TP and checkpoint costs; these curves are not a total-runtime comparison.

[E1] Saved trajectories; solid = channel 0, dashed = channel 1. No optimizer rerun for these figures.

FRAMES is close to the gate, but the comparison is unfinished

Paired two-qubit channels 0/1 have byte-identical data and initial factors across the archives below.

Method	Channel	Updates	RMSE	Normalized TP	Joint gate
Printed ratio[1]	0	100,000	1.045e-03	2.205e-03	FAIL
Printed ratio[1]	1	100,000	1.753e-03	3.417e-03	FAIL
FRAMES[2]	0	400,000	1.111e-06	9.523e-06	FAIL
FRAMES[2]	1	400,000	8.427e-07	7.953e-06	PASS
Armijo FGD[1]	0	1,747	1.326e-10	2.363e-10	PASS
Armijo FGD[1]	1	2,218	1.436e-10	2.656e-10	PASS
Exact-ray FGD[3]	0	1,401	8.928e-11	3.473e-10	PASS
Exact-ray FGD[3]	1	1,882	1.004e-10	3.833e-10	PASS

[1] Fixed $\alpha=5\times10^{-4}$ ($\lambda=0.064$). [2] FRAMES $\alpha_0=5\times10^{-5}$, $\alpha(t)=\alpha_0(t+1)^{0.25}$, batch 32, radius 2. [3] Fixed $\alpha=5\times10^{-5}$ ($\lambda=0.0064$). FGD runs are unconstrained in factor space; FRAMES uses its norm-ball domain.

FRAMES satisfies TP and fidelity on both channels; channel 0 misses RMSE by 11.1%. Printed ratio and FRAMES each charge 12.8 million nominal rows here, excluding spectral/TP costs. Both line-search controls reach much lower residuals with fewer charged rows; a final best-for-best wall-time ranking is not established.

[E2] Eight-run Gell-Mann diagnostic (7 October); [E1] selected exact-ray runs (8 October). Unequal budgets and step costs.

Higher qubits: preserve the evidence, qualify the claims

Earlier four-qubit recovery is real; the new frozen exact-ray configuration has only been tested at two qubits.

Earlier exploratory n=4 results · 8,192 rows · different initialization / tuning

Method / channel seeds	Gate	RMSE range	Budget / TP policy
Armijo FGD / 0–3	4/4	1.52e−7 – 8.66e−7	20k updates; fixed λ=0.4096
Printed ratio / 1–3	0/3	4.56e−3 – 4.77e−3	2k updates; fixed λ=40.96
Early FRAMES / 1–3	0/3	2.94e−3 – 3.26e−3	512k updates; CONSTANT penalty

This establishes recoverability at n=4, not a fair speed ranking: Armijo used ~459–460 million charged rows versus 16.384 million for each other run. Early constant-penalty FRAMES is not the intended increasing-penalty comparator. These old channels are no longer held out.

Legacy n=8 remains useful—but is a separate benchmark

A verified local-Pauli continuation reached 99.14% factor fidelity after 200k updates, with Gaussian noise σ=0.05 and normalized TP 0.0107. This is not a current strict-gate certificate or a result on the corrected global-probe design.

Also correct the old memory claim: 64 GiB at n=8 is the OPTIONAL dense adaptive-numerator matrix. Current code has a matrix-free spectral eigensolve and exact factorized TP. Full passes remain costly; large-n superiority is still unestablished.

[E3] Independently checked n=4 historical endpoints; [E4] matrix-free code/tests; [E5] verified legacy n=8 snapshot.

What we can present—and what we should do next

Suggested conclusion: a working fixed-penalty FGD baseline, with exact-paper reproduction and final comparison still open.

Supported conclusions

1. The corrected rank-one inverse problem supports accurate recovery: modified exact-ray FGD at n=2, and earlier Armijo FGD at n=4.
2. Step selection materially affects the observed outcome; failure of our printed-rule reconstruction does not refute Quiroga's paper.

Next experimental decisions

Freeze explicit method labels. Retain the printed ratio as a reproduction track and line-search FGD as a separate strong control. Tune FRAMES's initial penalty and continuation independently of FGD's fixed penalty; compare time-to-the-same-gate, memory and complete operator work. Confirm at n=3/4 before scaling further, then evaluate a genuinely untouched test set.

Checks and evidence

Latest batch: 64 archive pairs, 16 complete task inventories, three pinned plans; independent endpoint, source, initialization and work-accounting checks. Recorded suite: 597 passed, 55 skipped; 16 additional harness tests. Latest source 4ce1108 is merged locally, not pushed. The new frozen setting has no noisy-data, higher-rank or higher-qubit validation yet.

[E1] Latest audits: screen / confirmation / historical revalidation; Ruche 2039775 / 2039794 / 2039814.

[E2] Gell-Mann diagnostic audit, source 966ef74; Ruche 2038514. [E3] n=4 source d62fcf1, 2–3 October.

[E4] qpt_quiroga_sensing.md and norm tests. [E5] n8_continue_200k_20260923_01/verification.json. Hashes attached.

[P1] Quiroga & Kyrillidis (2023), §III-B and §V

No new experiments were submitted for this briefing. All earlier results and failures are retained.